

Solutions to JEE Advanced – 3 | JEE 2022 | Paper-1

PHYSICS

- 1.(D) Going from A towards B,

$$-2(3) - 8 - 2(1) - 2(2) + E - 2(1) = 0 \Rightarrow E = 22 \text{ V}$$

- 2.(B) Immediately after the switch is closed, the capacitors behave like zero resistance wires, and hence all three resistances are in parallel, giving a net resistance $\frac{R}{3}$

Therefore, immediately after the switch is closed, the current through it is $\frac{V}{\left(\frac{R}{3}\right)} = \frac{3V}{R}$

After a long time, the current through both capacitor branches is zero, and hence they can be removed from the circuit for analysis. So, now the three resistances are in series, and hence the current through the switch is $\frac{V}{3R}$

- 3.(A) Let the flux through the surface A due to the negative and positive charge be respectively Φ_A^- and Φ_A^+ .

Let the flux through the surface B due to the negative and positive charge be respectively Φ_B^- and Φ_B^+ .

Then, because the surface A is closer to the negative charge than the positive charge, $|\Phi_A^-| > |\Phi_A^+|$

And, because the surface B is closer to the positive charge than the negative charge, $|\Phi_B^-| < |\Phi_B^+|$

Since it is clear already that $\Phi_A^- > 0$ and $\Phi_A^+ < 0$, and that $\Phi_A = \Phi_A^- + \Phi_A^+$, we can now say that $\Phi_A > 0$.

Similarly, because it already clear that $\Phi_B^- < 0$ and $\Phi_B^+ > 0$, and that $\Phi_B = \Phi_B^- + \Phi_B^+$, we can say that $\Phi_B > 0$.

- 4.(A) Net capacitance of the combination of C and 2C in parallel is $C + 2C = 3C$

So, net capacitance between A and B is $\frac{(3C)(C)}{3C + C} = \frac{3C}{4}$

So, total charge that flows out of the battery is $Q = \frac{3CE}{4}$

So, potential difference across the combination of C and 2C in parallel is $\frac{Q}{3C} = \frac{E}{4}$

- 5.(D) Since the second ball is brought to the position initially occupied by the first ball, in the equilibrium condition, the compression in the spring is exactly equal to the separation between the balls
Let it be x

Then, balancing forces on the ball attached to the spring, $\frac{Q^2}{4\pi\epsilon_0 x^2} = kx \Rightarrow x = \left(\frac{Q^2}{4\pi\epsilon_0 k}\right)^{1/3}$

Work done against spring force, $W_1 = \frac{1}{2} kx^2 = \frac{1}{2} k \left(\frac{Q^2}{4\pi\epsilon_0 k}\right)^{2/3}$

Work done against electrostatic force = Change in electrostatic potential energy

$$= W_2 = \frac{Q^2}{4\pi\epsilon_0 x} - 0 = k \left(\frac{Q^2}{4\pi\epsilon_0 k}\right)^{2/3}$$

So, net work, $W = W_1 + W_2 = \frac{3}{2}k \left(\frac{Q^2}{4\pi\epsilon_0 k} \right)^{2/3} = \frac{3}{2} \left(\frac{1}{4\pi\epsilon_0} \right)^{2/3} Q^{4/3} k^{1/3}$

Therefore, $W \propto Q^{4/3} k^{1/3}$

6.(B) Taking a Gaussian surface of radius x (such that $\frac{R}{2} \leq x \leq R$), and applying Gauss law,

$$E(4\pi x^2) = \frac{Q_{encl}}{\epsilon_0}$$

Now, $Q_{encl} = \int_{R/2}^x \rho_0 \left(\frac{r}{R} \right) (4\pi r^2 dr) = \frac{\pi \rho_0}{R} \left(x^4 - \frac{R^4}{16} \right)$

Therefore, $E = \frac{\rho_0}{4\epsilon_0 R} \left(x^2 - \frac{R^4}{16x^2} \right)$

7.(ABCD)

(A) Applying Gauss Law on a spherical Gaussian surface of radius $2R$,

$$\Rightarrow E(4\pi(2R)^2) = \frac{-Q}{\epsilon_0} \Rightarrow E = -\frac{Q}{16\pi\epsilon_0 R^2}$$

This means the magnitude is $\frac{Q}{16\pi\epsilon_0 R^2}$ and the field is directed radially inward.

(B) We can observe easily using Gauss law that the electric field inside the inner shell is zero

(C) Potential at the common centre is the same as the potential of the inner shell

$$V_{centre} = \frac{-Q}{4\pi\epsilon_0 R} + \frac{Q}{4\pi\epsilon_0 (3R)} = -\frac{Q}{6\pi\epsilon_0 R}$$

(D) Potential of outer shell: $V_{outer} = \frac{Q_{total}}{4\pi\epsilon_0 (3R)} = \frac{-Q + Q}{4\pi\epsilon_0 (3R)} = \text{zero}$

8.(BC)

Let the charge on the plates be Q and $-Q$

Let the area of the plates be A each

Then, the electric field outside and inside the slab are respectively

$$E_1 = \frac{Q}{\epsilon_0 A} \quad \text{and} \quad E_2 = \frac{Q}{2\epsilon_0 A}$$

So, the potential difference between the plates, $E_1 \left(\frac{d}{2} \right) + E_2 \left(\frac{d}{2} \right) = \frac{3}{4} \frac{Qd}{\epsilon_0 A} = V$

Therefore, $Q = \frac{4}{3} \left(\frac{\epsilon_0 A}{d} \right) V$

So, the electric field inside the slab, $E_2 = \frac{Q}{2\epsilon_0 A} = \frac{2V}{3d}$

And, the potential energy stored in the capacitor, $U = \frac{1}{2} QV = \frac{2}{3} \left(\frac{\epsilon_0 A}{d} \right) V^2 = \frac{2}{3} CV^2$

9.(BD) Net electrostatic force on the floating ball is $F_E = 2 \left[\frac{q^2}{4\pi\epsilon_0 \left(\left(\frac{a}{2} \right)^2 + \left(\frac{\sqrt{3}a}{2} \right)^2 \right)^{3/2}} \left(\frac{\sqrt{3}a}{2} \right) \right] = \frac{\sqrt{3}q^2}{4\pi\epsilon_0 a^2}$

As the floating ball is in equilibrium, $mg = F_E = \frac{\sqrt{3}q^2}{4\pi\epsilon_0 a^2}$

After the vessel is filled with the liquid, the buoyant force on the floating ball, $B = \left(\frac{\rho}{\rho_0} \right) mg$

Balancing forces, $mg = B + \frac{F_E}{\kappa} \Rightarrow \kappa = \frac{\rho_0}{\rho_0 - \rho}$

10.(ACD)

(A) Let the potential at a point P on the circular base be V_P

If we imagine a symmetrical hemisphere below the given one, the potential at P due to that hemisphere is also V_P

Therefore, the net potential at P due to the complete sphere created is $V_P + V_P = 2V_P$

But since the sphere is uniformly charged over its surface, the entire volume inside it has no electric field and is therefore equipotential. So, the potential of the point P is independent of its location.

However, we must note carefully that this only works if the point P is on the circular base of the original hemisphere, and does not work if P is anywhere else inside the hemisphere, because then the potential at P will not be equal due to each hemispherical part.

Therefore, the circular base of the original hemisphere is an equipotential surface.

(B) It can be seen by dividing the hemisphere into ring elements parallel to the circular base that the electric field at O due to each such element points perpendicular to the base. Hence, the net field at O also points perpendicular to the base and is not zero.

(C) Since each point of the hemisphere is at a distance R from O , the potential at O ,

$$V_O = \frac{Q}{4\pi\epsilon_0 R} = \frac{\sigma(2\pi R^2)}{4\pi\epsilon_0 R} = \frac{\sigma R}{2\epsilon_0}$$

So, the work done in bringing a charge $+q$ from infinity to O ,

$$W = qV_O = \frac{\sigma R q}{2\epsilon_0}$$

(D) Again, if we add a symmetrical hemispherical below the given hemisphere to complete a sphere charged uniformly over its surface with charge density σ , we can see that the electric potential at a point (say point M) in the same plane as the circular base, and at a distance $2R$ from O is

$$V_M = \frac{Q}{4\pi\epsilon_0 (2R)} = \frac{\sigma(4\pi R^2)}{8\pi\epsilon_0 R} = \frac{\sigma R}{2\epsilon_0}$$

But, due to symmetry, the potential at M due to each hemisphere is equal. Hence, the potential at M

due to just the original hemisphere is $\left(\frac{1}{2} \right) \frac{\sigma R}{2\epsilon_0} = \frac{\sigma R}{4\epsilon_0}$

11.(ABCD)

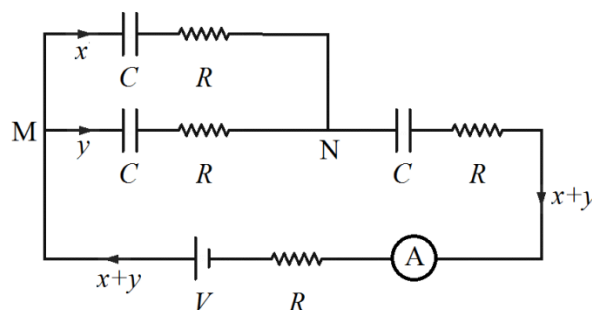
Initially the capacitors are uncharged and hence the potential difference across them is zero. So, immediately after S1 is closed, the current through the ammeter is $I_1 = \frac{V}{4R}$.

With S1 closed and S2 open, steady state is reached, and the potential difference across each of the capacitors in series with the battery becomes $\frac{V}{2}$ and charge on each of them becomes $\frac{CV}{2}$.

So, work done by the battery until now, $W_1 = \left(\frac{CV}{2}\right)V = \frac{1}{2}CV^2$

When S2 is closed, the charge on these capacitors cannot change immediately and hence the potential difference across them also cannot change immediately.

We can assume the current that flows in each branch immediately after S2 is closed as follows:



Applying KVL in the smaller loop: $xR - \frac{V}{2} - yR = 0 \Rightarrow x - y = \frac{V}{2R}$

Applying KVL in the larger loop:

$$(x+y)R + \frac{V}{2} + yR + \frac{V}{2} - V + (x+y)(2R) = 0 \Rightarrow 3x + 4y = 0$$

Solving the equations, we get $x = \frac{2V}{7R}$ and $y = -\frac{3V}{14R}$

So, the current through the ammeter is $x + y = \frac{V}{14R}$

Now, a long time after S2 is closed, the current in every branch is zero and the potential difference across the capacitors in parallel is equal. Since their capacitance is same, this means the charge on them must be equal too. Let this charge be Q_1 and let both their left plates be positively charged.

Let the charge on the third capacitor be Q_2 and let its left plate be positively charged

From KVL in the bigger loop, $V - \frac{Q_1}{C} - \frac{Q_2}{C} = 0 \Rightarrow Q_1 + Q_2 = CV$

Also, the right plates of the capacitors in parallel, and the left plate of the third capacitor, all three form an isolated system. So, from conservation of charge on these plates (between the very initial condition when S1 and S2 are open and the final condition),

$$0 = -Q_1 - Q_1 + Q_2 \Rightarrow Q_2 = 2Q_1$$

Solving, we get $Q_1 = \frac{CV}{3}$ and $Q_2 = \frac{2CV}{3}$

So, the total charge that flowed out of the junction M after S2 was closed, $Q = Q_1 + \left(Q_1 - \frac{CV}{2}\right) = \frac{CV}{6}$

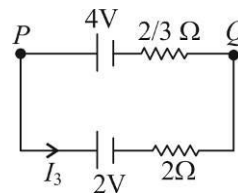
This is exactly the amount of charge that must have flowed out of the positive terminal of the battery over a long time after S2 was closed.

Therefore, the work done by the battery over a long time after S2 is closed, $W_2 = \left(\frac{CV}{6}\right)V = \frac{1}{6}CV^2$

- 12.(ACD) The equivalent battery of E_1 and E_2 has E_{eq} and r_{eq} given by

$$E_{eq} = \frac{\frac{E_1 + E_2}{\frac{1}{r_1} + \frac{1}{r_2}}}{\frac{1}{r_1} + \frac{1}{r_2}} = \frac{\frac{4 + 4}{\frac{1}{1} + \frac{1}{2}}}{\frac{1}{1} + \frac{1}{2}} = \frac{6}{\left(\frac{3}{2}\right)} = 4V$$

$$\frac{1}{r_{eq}} = \frac{1}{r_1} + \frac{1}{r_2} = \frac{1}{1} + \frac{1}{2} = \frac{3}{2} \Rightarrow r_{eq} = \frac{2}{3}\Omega$$



The circuit can be drawn as $I_3 = \frac{4-2}{\frac{2}{3} + 2} = 2 \times \frac{2}{8} = 0.75 A$; $V_P - V_Q = 4 - (0.75)\left(\frac{2}{3}\right) = 3.5 V$

For the uppermost branch of the original circuit:

$$V_P - V_Q = E_1 - I_1 r_1 ; 3.5 = 4 - I_1(1) \Rightarrow I_1 = 0.5 A ; I_2 = I_3 - I_1 = 0.75 - 0.5 = 0.25 A$$

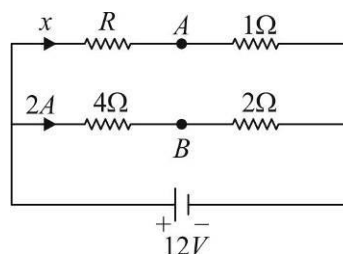
- 13.(3) Since the voltmeter has infinite resistance there will be no current through it. Hence, we can remove it from the circuit before our analysis. The currents in the two branches will be as shown:

The current $x = \frac{12}{R+1}$

We are given that $V_A - V_B = -1$
Therefore, $+2(4) - xR = -1$

$$\Rightarrow 8 - \frac{12R}{R+1} = -1$$

Solving, we get $R = 3\Omega$.



- 14.(0.33) Let us assume the positive terminal of the battery was connected to the point M

Equivalent capacitance of branch MNO is $\frac{(2C)(C)}{2C + C} = \frac{2C}{3}$

Therefore, total charge that flows into branch MNO until steady state is $\frac{2CE}{3}$

So, at steady state, $V_M - V_N = \frac{\left(\frac{2CE}{3}\right)}{2C} = \frac{E}{3}$

Similarly, we can find that at steady state, $V_M - V_P = \frac{2E}{3}$

Therefore, we get $V_N - V_P = \frac{E}{3}$

- 15.(0.40) Since the instantaneous potential difference across both resistances is always equal (and equal to the instantaneous potential difference across the capacitor, say V), the rate of heat dissipation in the resistances is:

$$P_1 = \frac{V^2}{R} \quad \text{and} \quad P_2 = \frac{V^2}{4R}$$

Since the instantaneous rates of heat dissipation in the resistances are always in the ratio 4:1, the total heat dissipated in them until the capacitor is completely discharged will also be in the ratio 4:1.

And the total heat dissipated in the resistances will be equal to the initial potential energy stored in the capacitor, $\frac{1}{2}CV_0^2$

Therefore, total heat dissipated in the R resistance, $H_2 = \frac{4}{5}\left(\frac{1}{2}CV_0^2\right) = (0.40)CV_0^2$

16.(2) With the particle at the centre of ring A (i.e. the initial situation), its potential energy is

$$U_I = \left(\frac{Qq}{4\pi\epsilon_0 R} \right) + \left(-\frac{Qq}{4\pi\epsilon_0 \sqrt{R^2 + (\sqrt{3}R)^2}} \right) = \frac{Qq}{8\pi\epsilon_0 R}$$

With the particle at the centre of ring B (i.e. the final situation), its potential energy is

$$U_F = \left(\frac{Qq}{4\pi\epsilon_0 \sqrt{R^2 + (\sqrt{3}R)^2}} \right) + \left(-\frac{Qq}{4\pi\epsilon_0 R} \right) = -\frac{Qq}{8\pi\epsilon_0 R}$$

Conserving energy,

Gain in kinetic energy = Loss in potential energy

$$\frac{1}{2}mv^2 - 0 = U_I - U_F = \frac{Qq}{8\pi\epsilon_0 R} - \left(-\frac{Qq}{8\pi\epsilon_0 R} \right) = \frac{Qq}{4\pi\epsilon_0 R} \Rightarrow v = \left(\frac{Qq}{2\pi\epsilon_0 R m} \right)^{1/2}$$

17.(26) Resistance of each bulb, $R = \frac{40^2}{4} = 400 \Omega$

Let the number of connected bulbs be n

$$\text{Then, the current through the battery, } i = \frac{60}{8 + \frac{400}{n}} = \frac{15n}{2n + 100}$$

$$\text{Hence, the potential difference across each bulb, } V = i \left(\frac{400}{n} \right) = \frac{3000}{(n + 50)}$$

$$\text{Therefore, the power consumed by each bulb, } P = \frac{V^2}{400} = \frac{22500}{(n + 50)^2}$$

$$\text{Now, } P < 4 \Rightarrow (n + 50)^2 > \frac{22500}{4} \Rightarrow n > 25$$

18.(6) The total charge on the charged spherical shell, $Q = (2\rho) \left(\frac{4\pi}{3} \left(\frac{R}{2} \right)^3 \right) + \rho \left(\frac{4\pi}{3} \left(R^3 - \left(\frac{R}{2} \right)^3 \right) \right) = \frac{3\pi\rho R^3}{2}$

For points outside it, the charged spherical shell will behave like a point charge Q fixed at the origin. We can find the work done by finding the initial and final potential energy of the system.

The initial situation is that a point charge Q is kept on the axis of the dipole at a distance $\frac{3R}{2}$ from it. So, for writing the potential energy, we can use the result for the electric potential at a point on the axis of a dipole. We note that the point charge Q is closer to the negative charge of the dipole, so the potential at the location of Q will be negative.

Similarly, the final condition can be seen as a point charge Q kept on the equator of a dipole, and we can use the result for the electric potential on the equator of a dipole to find the final potential energy.

$$\text{Therefore, the initial potential energy, } U_1 = Q \left(-\frac{p}{4\pi\epsilon_0 \left(\frac{3R}{2} \right)^2} \right) = -\frac{pQ}{9\pi\epsilon_0 R^2}$$

And, the final potential energy, $U_2 = Q(0) = 0$

$$\text{So, the work done, } W = U_2 - U_1 = \frac{pQ}{9\pi\epsilon_0 R^2} = \left(\frac{p}{9\pi\epsilon_0 R^2} \right) \left(\frac{3\pi\rho R^3}{2} \right) = \frac{1}{6} \frac{p\rho R}{\epsilon_0}$$

CHEMISTRY

- 1.(D) As $k_1 > k_2$ then concentration of Y will increase at faster rate than concentration of Z.
Thus Y will achieve a maxima of its concentration after which only we will observe any significant rise in concentration of Z.
- 2.(C) A binary mixture of ethanol and propanol is an ideal solution. Due to higher molecular mass $P_{\text{propanol}}^\circ < P_{\text{ethanol}}^\circ$. Figure (C) is most appropriate for fractional distillation of that mixture.
- 3.(D) Diamond-cubic is a F.C.C arrangement of carbon atoms. [A], [B] and [C] are correctly matched.



$$t = 0 \quad \quad \quad P_i \quad \quad \quad 0 \quad \quad \quad 0$$

$$t = t_{100\%} \quad \quad \quad 0 \quad \quad \quad 2P_i \quad \quad \quad P_i$$

$$P_{\text{Total}} \text{ at } t_{100\%} = 3P_i = 360 \text{ torr} \Rightarrow P_i = 120 \text{ torr}$$

$$\text{Total pressure after any time } t, P'_T = (P_i - P) + 3P = P_i + 2P$$

$$\text{As } (P_i + 2P) = 240 \Rightarrow 2P = 240 - 120 = 120 \Rightarrow P = 60 \text{ torr}$$

$$\therefore 10 \text{ min} = \text{Half life of the reaction} = t_{1/2} \text{ and } 30 \text{ min} = 3t_{1/2}$$

$$\text{Pressure of } (\text{CH}_3)_3\text{C}-\text{O}-\text{O}-\text{C}(\text{CH}_3)_3 \text{ after } 3t_{1/2} = \frac{P_i}{8} = 15 \text{ torr}$$

$$\text{Rate of disappearance of } (\text{CH}_3)_3\text{C}-\text{O}-\text{O}-\text{C}(\text{CH}_3)_3 = k(P_i) = \frac{0.693}{10}[15] = 1.5[0.693] = 1.04 \text{ torr / min}$$

5.(B) As $\frac{P^\circ - P_s}{P_s} = \frac{mM_{(\text{solvent})}^\circ}{1000}$

$$\frac{80 - 75}{75} = \frac{m[18]}{1000} \Rightarrow m = \frac{5 \times 1000}{75 \times 18} = 3.7 \text{ molal}$$

$$\Delta T_f = K_f m = 1.86 \times 3.7 = 6.882^\circ\text{C}$$

$$T_f = -6.882^\circ\text{C}$$

$\therefore$ Option B is correct

- 6.(A) That smaller cube will have tetrahedral voids at its corners and octahedral void at its center, edge length of the

$$\text{smaller cube} = \frac{a}{2} = \sqrt{2}R$$

$$\text{Radius of the tetrahedral void} = 0.225R$$

$$\text{Radius of the octahedral void} = 0.414R$$

$$\text{Packing fraction} = \frac{\frac{4}{3}\pi[0.225R]^3 + \frac{4}{3}\pi[0.414R]^3}{(\sqrt{2})^3 R^3} \times 100 = 12.22\%$$

$$\left[\begin{array}{ll} \text{Put } (0.225)^3 = 0.0114 & (0.414)^3 = 2.83 \\ (0.414)^3 = 0.0710 & (\pi/3) = 1.05 \end{array} \right]$$

7.(ACD) For the first step $A \xrightleftharpoons[k_2]{k_1} B$ at a given temperature $k_1 < k_2$

Thus $E_1 > E_2$ (Activation energies)

$$\therefore \Delta H_{(1^{st})} = E_1 - E_2 = (+)ve$$

Hence conversion of A to B is an endothermic process

For second step $B \xrightleftharpoons[k_4]{k_3} C$ at a given temperature $k_3 > k_4$

Thus $E_3 < E_4$ (Activation energies)

$$\therefore \Delta H_{(2^{nd})} = E_3 - E_4 = (-)ve$$

Hence conversion of B to C is an exothermic process

Now $A \xleftarrow{k_2} B \xrightarrow{k_3} C$ at a given temperature $k_2 < k_3$

Thus $E_2 > E_3$

Hence activation energy for conversion of $B \longrightarrow A$ is higher than that for $B \longrightarrow C$

$k_1 < k_3$; hence $A \longrightarrow B$ is the slow or rate determining step. Overall order of the reaction is 1.

8.(A) Colligative properties are driven by entropy factor and dependent of amount of the solution. Colligative properties increases by ionization of the solute dissolved in water.

9.(ABC) In FCC tetrahedral and octahedral voids are present.

10.(ABCD)

$$K = \frac{2.303}{t} \log_{10} \frac{r_{\infty} - r_0}{r_{\infty} - r_t}$$

$$r_0 = +32^\circ; \quad r_{\infty} = -12^\circ; \quad r_t = +20^\circ$$

$$t = 50 \text{ min}$$

$$K = \frac{2.303}{50} \log_{10} \frac{44}{32} = \frac{2.303}{50} \log_{10} \frac{11}{8}$$

$$K = \frac{2.303}{50} [1.05 - 0.90]$$

$$K = \frac{2.303}{50} [0.15]$$

$$t_{1/2} = \frac{2.303}{\frac{2.303 \times 0.15}{50}} \log_{10} 2 = \frac{50 \times 0.30}{0.15} = 100 \text{ min}$$

To get the time when the solution become optically inactive [put $r_t = 0$]

$$\frac{2.303[0.15]}{50} = \frac{2.303}{t'} \log \frac{44}{12}$$

$$\Rightarrow t' = \frac{50}{0.15} \log \frac{11}{3} = \frac{50}{0.15} [1.05 - 0.48] = 190 \text{ min}$$

If after 190 minutes solution is optically inactive so at $t = t_{1/2} = 100$ minutes solution must be dextrorotatory.

11.(AD) For benzoic acid in water

$$\Delta T_B = (im)K_B$$

$$(im) = \frac{1}{0.83}$$

$$\Delta T_f = \left(\frac{1}{0.83}\right)1.86 = 2.24^\circ\text{C} \Rightarrow T_f = -2.24^\circ\text{C}$$

As i for benzoic acid is different in water and in benzene hence boiling point can't be calculated.

As in water benzoic acid undergoes ionisation while in benzene it undergoes dimerisation thus observed molar mass of the benzoic acid in water is lower than 122 g/mole while in benzene it is greater than 122 g/mole.

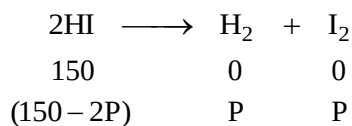
12.(BC)

(A) Frenkel and Schottky defects are stoichiometric defects

Metal deficiency or metal excess defects can cause increase or decrease in density and increase in electrical conductivity of the ionic solids

Statement (D) is correct

13.(150) According to stoichiometry of the reaction, total pressure will remain constant



$$P_{\text{Total}} = (150 - 2P) + 2P = 150 \text{ torr}$$

14.(128) As $\Delta T_B = k_B m$

$$m = \frac{50/80}{2} = \frac{5}{16} \text{ molal}$$

$$k_B = \frac{RT_B^2 M_{\text{solvent}}}{1000[\Delta H_{\text{vap}}]} = \frac{8 \times 400 \times 400 \times 100}{1000 \times 40 \times 1000} = 3.2 \quad \therefore \Delta T_B = 3.2 \left[\frac{5}{16} \right] = 1 \Rightarrow T_B = 128^\circ\text{C}$$

15.(32) In diamond cubic structure effectively 8 constituent particles are present per unit cell, four from ccp lattice positions and four from alternate tetrahedral voids. In diamond cubic structure, carbon occupies tetrahedral voids hence its coordination number is four.

16.(75)



$$\begin{array}{ccc} P_i & 0 & 0 \\ P_i - x & 2x & x/2 \end{array}$$

$$P_T = (P_i + 1.5x) = 380 \Rightarrow (P_i - x) = 80 \Rightarrow x = 120$$

$$k = \frac{2.303}{100} \log \left(\frac{200}{80} \right) \Rightarrow k = \frac{2.303}{100} \log \left(\frac{5}{2} \right) \Rightarrow k = \frac{2.303}{100} [0.70 - 0.30] \text{ or } k = \frac{2.303}{100} [0.40]$$

$$\frac{2.303}{t_{1/2}} \log_{10} 2 = \frac{2.303}{100} [0.40]$$

$$t_{1/2} = \frac{100 \times 0.30}{0.40} = 75 \text{ min}$$



$$0.1 \text{ M} \quad 0.2 \text{ M}$$

$$200 \text{ mL} \quad V \text{ mL}$$

[20 m moles]

$$\Rightarrow \text{Volume of NaOH added} = 100 \text{ mL}$$

$$\text{Final volume (before dilution)} = 300 \text{ mL}$$

$$\text{Final volume (after dilution)} = 300 + 1700 = 2000 \text{ mL}$$

$$[\text{NaCl}] = \frac{20 \text{ m moles}}{2000 \text{ mL}} = 10^{-2} \text{ M}$$

$$\text{Vant Hoff factor for NaCl} = 1 + \alpha = 1.98$$

$$\text{Osmotic pressure} = [\text{NaCl}]iRT = 10^{-2}[1.98][0.0821][308] = 0.500$$

18.(18.75)

For Ti_{2x}O_3

$$\text{Total positive charge} = 6x$$

$$\text{Total negative charge} = 6$$

$$\text{Extra positive charge required} = 6 - 6x$$

$$\text{Number of } \text{Ti}^{+4} = 6 - 6x$$

$$\text{Number of } \text{Ti}^{+3} = 2x - [6 - 6x] = 8x - 6$$

$$\frac{\text{Ti}^{+4}}{\text{Ti}^{+3}} = \frac{6 - 6x}{8x - 6} = \frac{1}{24}$$

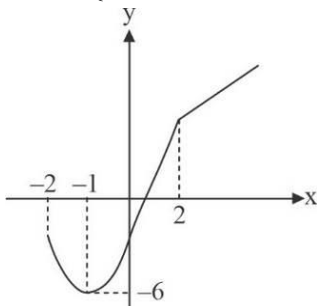
$$(6 - 6x)24 = (8x - 6) \Rightarrow (6 - 6x)12 = [4x - 3] \Rightarrow 72 - 72x = 4x - 3$$

$$\Rightarrow 76x = 75 \Rightarrow x = \frac{75}{76}$$

$$\therefore 19x = \frac{75 \times 19}{76} = 18.75$$

MATHEMATICS

$$1.(C) \quad f(x) = \begin{cases} 2x^2 + 4x - 4 & x \in [-2, 2] \\ 4x + 4 & x \in (2, \infty) \end{cases}$$



$\therefore$ Range : $[-6, \infty)$

$$2.(C) \quad \text{Let a point on } y^3 = x^4 \text{ be } (t^3, t^4)$$

$$3y^2 y' = 4x^3 \Rightarrow y' = \frac{4x^3}{3y^2} \Rightarrow y' = \frac{4}{3}t$$

$$\text{Equation of tangent is } y - t^4 = \frac{4t}{3}(x - t^3)$$

$\therefore$ it is a normal to $x^2 + y^2 - 2x = 0$

$\therefore$ it must pass through $(1, 0)$

$$\Rightarrow -\frac{3}{4}t^3 = 1 - t^3 \Rightarrow \frac{t^3}{4} = 1 \Rightarrow t^3 = 4$$

$$\text{Now } m = \frac{4t}{3} \Rightarrow \left(\frac{3m}{4}\right)^3 = t^3 = 4$$

$$3.(B) \quad 2f(x) \cdot f'(x) - 2f'(x) - \frac{1}{2\sqrt{x}} = 0$$

$$2(f(x) - 1)f'(x) = \frac{1}{2\sqrt{x}}$$

Integrating both the side

$$(f(x) - 1)^2 - \sqrt{x} = C$$

$$\therefore f(0) = 2; \quad c = 1$$

$$\Rightarrow (f(x) - 1)^2 = \sqrt{x} + 1 \Rightarrow f(x) = 1 + \sqrt{\sqrt{x} + 1}$$

$$\text{Now, } \lim_{x \rightarrow 0^+} \frac{f(x) - 2}{\sqrt{x}} = \lim_{x \rightarrow 0^+} \frac{\sqrt{\sqrt{x} + 1} - 1}{\sqrt{x}}$$

$$\lim_{x \rightarrow 0^+} \frac{\sqrt{x} + 1 - 1}{\sqrt{x}(\sqrt{\sqrt{x} + 1} + 1)} = \frac{1}{2}$$

$$4.(C) \quad \therefore g'(y) = \frac{1}{f'(x)} \Rightarrow g'(4) = \frac{1}{f'(64)} \quad (\because f(64) = 4)$$

$$= \frac{1}{\frac{1}{96}} = 96$$

5.(B) $p'(x) = x(2a_1 + 4a_2x^2 + \dots)$
 $= +ve \times x$
 $\frac{-}{+}$
 $\therefore$ exactly one minimum

6.(A) $2x^2 + y^2 = 12$ or $\frac{dy}{dx} = -\frac{2x}{y}$. Slope of normal at point $A(2, 2)$ is $\frac{1}{2}$. Also, point $B\left(-\frac{22}{9}, -\frac{2}{9}\right)$ lies on the curve and slope of AB is $\frac{2 - (-2/9)}{2 - (-22/9)} = \frac{1}{2}$

Hence, the normal meets the curve again at point $\left(-\frac{22}{9}, -\frac{2}{9}\right)$

7.(CD) $y = \sqrt{4x^2 - 4x + 2} + |x| = \sqrt{(2x-1)^2 + 1} + |x|$
 if $x < 0$

$$y = \sqrt{(2x-1)^2 + 1} - x$$

$$y' = \frac{(2x-1) \cdot 2}{\sqrt{(2x-1)^2 + 1}} - 1$$

$$y' < 0 \forall x < 0$$

$$\text{if } x > 0$$

$$y = \sqrt{(2x-1)^2 + 1} + x$$

$$y' = \frac{2(2x-1)}{\sqrt{(2x-1)^2 + 1}} + 1 \Rightarrow y' > 0 \Rightarrow \frac{\sqrt{3}-1}{2\sqrt{3}} < x < \infty ; y' < 0 \Rightarrow 0 < x < \frac{\sqrt{3}-1}{2\sqrt{3}}$$

8.(ACD)

$$f(x) = \log_a(4ax - x^2) \quad \left[\frac{3}{2}, 2\right]$$

$$a > 1 \Rightarrow f(x) = 4ax - x^2$$

$$\text{must be } \uparrow \forall x \in \left[\frac{3}{2}, 2\right]$$

$$\Rightarrow 2a \geq 2 \Rightarrow a \geq 1$$

$$\Rightarrow a > 1$$

$$0 < a < 1$$

$$\Rightarrow f(x) = 4ax - x^2$$

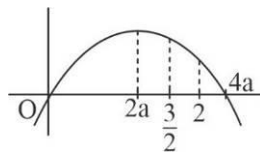
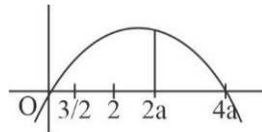
$$\text{must be } \downarrow \forall x \in \left[\frac{3}{2}, 2\right]$$

$$2a \leq \frac{3}{2} \text{ and } 4a > 2$$

$$a \leq \frac{3}{4} \text{ and } a > \frac{1}{2}$$

$$\frac{1}{2} < a \leq \frac{3}{4}$$

$$\left(\frac{1}{2}, \frac{3}{4}\right] \cup (1, \infty)$$



9(BD) See theory of method – 2 (Second derivative test)

10.(CD) At $x=0$ & $\frac{\pi}{2}$, $f(x)$ is discontinuous

At $x = 0$ for LHL $x = 0 - h$

$$\lim_{h \rightarrow 0} \left[\tan^2 \left(-h - \frac{\pi}{4} \right) \right] = 1$$

$$\text{RHL } \lim_{h \rightarrow 0} \left[\tan^2 \left(h - \frac{\pi}{4} \right) \right] = 0$$

Similarly at $x = \frac{\pi}{2}$ In between it is fine.

11.(AC) We have $\int \frac{\sin x - \cos x}{(\sin x + \cos x) \sqrt{\sin x \cos x + \sin^2 x \cos^2 x}} dx = \operatorname{cosec}^{-1}(g(x)) + c$

$$I = \int \frac{-\cos 2x}{(1 + \sin 2x) \sqrt{\sin x \cos x + \sin^2 x \cos^2 x}} dx$$

$$\text{Let } 1 + \sin 2x = y \Rightarrow I = \int \frac{-1 dy}{y \sqrt{y^2 - 1}} \Rightarrow I = \operatorname{cosec}^{-1}(y) + c$$

12.(ABCD) Since $f(x)$ is even, so we check in $[0, 2]$

$$f(x) = \begin{cases} ax^2 + b, & 0 \leq x < 1 \\ 1, & x = 1 \\ \frac{\lambda}{x}, & 1 < x \leq 2 \end{cases}$$

For continuity, $a + b = 1 = \lambda$ (1)

$$f'(x) = \begin{cases} 2ax, & 0 < x < 1 \\ -\frac{\lambda}{x^2}, & 1 < x < 2 \end{cases} \Rightarrow \text{For continuity, } 2a = -\lambda \quad \dots\dots\dots(2)$$

From (1) and (2), $a = -\frac{1}{2}$, $b = \frac{3}{2}$, $\lambda = 1$

13.(5.67) $a = 3$, $b = 12$, $c = 9$

$$\lim_{x \rightarrow 0} \frac{ax \left(1 + x + \frac{x^2}{2} + \frac{x^3}{6} \dots \right)}{x^2 \left(x - \frac{x^3}{6} \right)} - b \left(x - \frac{x^2}{2} + \frac{x^3}{3} \dots \right) + c.x \left(1 - x + \frac{x^2}{2} \dots \right) \frac{1}{x^2 \left(x - \frac{x^3}{6} + \dots \right)} = 2$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{x(a-b+c) + x^2\left(a + \frac{b}{2} - c\right) + x^3\left(\frac{a}{2} - \frac{b}{3} + \frac{c}{2}\right) + \dots}{x^3} = 2$$

$$a-b+c=0 \quad a + \frac{b}{2} - c = 0$$

$$\frac{a}{2} - \frac{b}{3} + \frac{c}{2} = 2 \quad \text{Solving we get } a=3, b=12, c=9$$

14.(20) e^{-1}

$$\begin{aligned} \text{Limit} &= \lim_{x \rightarrow 0} \left(1 - \frac{x - \sin x}{x} \right)^{\frac{\sin x}{x - \sin x}} \\ &= \lim_{x \rightarrow 0} \left\{ \left(1 + \frac{-1}{\left(\frac{x}{x - \sin x} \right)} \right)^{\frac{\frac{x}{x - \sin x}}{\frac{\sin x}{x}}} \right\} = \left\{ \lim_{y \rightarrow \infty} \left(1 + \frac{-1}{y} \right)^y \right\}^{\lim_{x \rightarrow 0} \frac{\sin x}{x}} \\ &\quad \left\{ \text{putting } \frac{x}{x - \sin x} = y \rightarrow \infty \text{ as } x \rightarrow 0 \right\} \\ &= (e^{-1})^1 = e^{-1} . \end{aligned}$$

15.(37) Given $f(2x+1) = 4x^2 + 14x$ (i)

We are find the sum of squares of roots of equation $f(x) = 0$, Putting $2x+1 = y$

$$\Rightarrow x = \frac{y-1}{2} \text{ in (i)}$$

$$\text{We have } f(y) = 4\left(\frac{y-1}{2}\right)^2 + 14\left(\frac{y-1}{2}\right)$$

$$f(y) = y^2 + 1 - 2y + 7y - 7 \Rightarrow f(y) = y^2 + 5y - 6$$

$$\therefore f(x) = 0 \quad \Rightarrow \quad x = -6 \text{ or } x = 1 \quad \therefore \alpha^2 + \beta^2 = 36 + 1 = 37$$

16.(-1) $\frac{dy}{dx} = \frac{a}{x} + 2bx + 1$

$$\text{at } x = -1, \frac{dy}{dx} = 0$$

$$\Rightarrow -a - 2b + 1 = 0 \quad \dots (1)$$

$$\text{at } x = 2, \frac{dy}{dx} = 0$$

$$\Rightarrow \frac{a}{2} + 4b + 1 = 0 \quad \dots (2)$$

$$\Rightarrow a + 8b + 2 = 0 \quad \dots (2)$$

$$(1) + (2) \Rightarrow 6b + 3 = 0$$

$$\Rightarrow b = -\frac{1}{2}$$

Using in (1) $a = 2$

$$17.(12) \frac{dy}{dx} = 3x^2 - 6x(7-k) - 3(9-k^2)$$

$$\text{Both roots of } \frac{dy}{dx} = 0$$

+ve

$$\Rightarrow a.f(0) > 0$$

$$\Rightarrow 3.(-3).(9-K^2) > 0$$

$$\Rightarrow K < -3 \text{ or } K > 3 \quad \dots (i)$$

$$\text{AND } D \geq 0$$

$$\Rightarrow 6^2(K-7)^2 + 4 \times 3 \times 3 \times (9-K^2) \geq 0 \Rightarrow K^2 - 14K + 49 + 9 - K^2 \geq 0$$

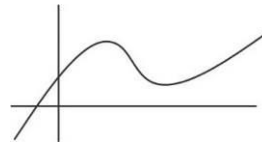
$$\Rightarrow 14K \leq 58 \Rightarrow K \leq \frac{29}{7} \quad \dots (2)$$

AND

$$\frac{-(-6)(7-K)}{2 \times 3} > 0$$

$$\Rightarrow K < 7 \quad \dots (3)$$

$$(1) \cap (2) \cap (3) \Rightarrow K < -3 \text{ or } K \in \left(3, \frac{29}{7}\right)$$



18.(3) LH Rule

$$\lim_{x \rightarrow 0} \frac{2f'(x) - 6f'(2x) + 4f'(4x)}{2x}$$

$$= \lim_{x \rightarrow 0} \frac{2f''(x) - 12f''(2x) + 16f''(4x)}{2}$$

$$= \frac{2p - 12p + 16p}{2} = \frac{6p}{2}$$

$$= 3p$$